Houssam Assaad

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Research Interests

Control, estimation and optimization for dynamic systems, applied so far to aerial, ground, and legged robots.

Education

American University of Beirut (AUB), Lebanon **Aug 2023 – May 2027 (expected)**
B.E. Mechanical Engineering, Minor in Economics; USAID HES Scholar *Beirut, Lebanon*

- **Relevant coursework:** System Analysis and Design (graduate, Modern Control), Control Systems, Design of Thermal Systems (Optimization), Instrumentation, Dynamics, Fluid Mechanics, Mechanical Design, Product Design and Development, Manufacturing Processes

Research Experience

King Fahd University of Petroleum & Minerals (KFUPM) **Jun 2026 – Sep 2026**
Research Intern, Control & Instrumentation Eng., supervisor Dr. Ramy Rashad *Dhahran, Saudi Arabia*

- Integrated a complete GNSS-denied sensing and estimation stack for a UAV: NVIDIA Jetson Orin Nano, Pixhawk 6X Pro (PX4) and Intel RealSense D435i, running ROS 2 Humble over a uXRCE-DDS bridge.
- Calibrated the IMU intrinsically, cutting gravity-magnitude error from 0.069 to 0.002 m/s^2, and characterized its noise with a 3-hour Allan-variance study (gyroscope bias instability 1.7–4.4 °/h).
- Benchmarked stereo-inertial SLAM (RTAB-Map) against OptiTrack motion capture over three trajectories: **0.100 m ATE RMSE**, path length within 6.8%; established a reproducible benchmarking protocol.
- First-author paper under review at the *Journal of Undergraduate Research International* (KFUPM).

American University of Beirut – Vision & Robotics Lab **Feb 2026 – Aug 2026**
*Undergraduate Research Assistant, supervisor Dr. Naseem Daher – **completed*** *Beirut, Lebanon*

- Formulated severity-aware pothole avoidance for autonomous ground vehicles as a Markov decision process; built discrete and continuous environments sharing one damage model.
- Developed tabular Q-learning, steering-only DQN and joint steering-and-speed DQN agents, with curriculum training and held-out test maps.
- Designed an exact-optimal evaluation baseline by backward dynamic programming over the finite acyclic state graph, defining a map-invariant *Normalized Damage Score* (NDS).
- DQN agent reached **NDS 88.8% ± 1.9%** on trained hazard densities and 51.4% on unseen densities (10 seeds), vs. 13.9% for the tabular baseline; the speed-extended agent modulates speed with hazard severity (NDS 83.6%).

Publications

H. Assaad, R. Rashad, "Benchmarking onboard visual–inertial SLAM against motion-capture ground truth," *Journal of Undergraduate Research International* (KFUPM), under review, 2026.

Experience

Middle East Airlines (MASCO) **Nov 2025 – Dec 2025**
Aircraft Maintenance Intern *Beirut, Lebanon*

- Trained on aircraft systems (avionics, airframe and powerplant) and toured maintenance facilities and workshops at Beirut Airport.

Projects

Autonomous Inspection Quadruped (Final Year Project) | *Unitree Go2, ROS 2, Nav2* **Sep 2026 – May 2027**

- Lead the inspection pipeline in a four-person team (advisor Dr. Hussein Hussein) building a quadruped that walks through a building, checks each element against the building's BIM model and reports what is missing or misplaced.
- Developing BIM-referenced LiDAR-inertial localization (adapting open-source LIO-BIM), a probabilistic semantic voxel map initialized from the BIM, and cost-aware navigation on Nav2.
- Integrating vision-language-model verification of inspected elements and coordinating interfaces with the teammates' camera arm and next-best-view planner. *(Ongoing.)*

Self-Balancing Robot: SISO to MIMO (MECH 436 & 653) | *Simulink, ESP32* **Nov 2025 – Present**

- **SISO (MECH 436):** designed and built a two-wheeled self-balancing robot, modeled it as an inverted pendulum and tuned a PID controller against sensor noise and actuator saturation; stabilized the hardware under applied load.
- **MIMO (MECH 653, graduate, ongoing):** extending to independent wheel torques that control pitch, forward velocity and heading; deriving a 6-state Lagrangian model and decoupling it into pitch and yaw subsystems.
- Identified a right-half-plane zero limiting speed tracking and loss of controllability under a single common input; designing pole-placement/LQR state feedback with observers on an ESP32, benchmarked against OptiTrack.

Heat Sink Fin-Array Optimization (MECH 510) | *MATLAB, optimization, neural networks* **Fall 2026**

- Formulated the design of a staggered oblong-fin heat sink as a constrained optimization in three geometric variables, maximizing fin effectiveness under a 10 Pa pressure-drop limit and an 8 mm manufacturing constraint.
- Calibrated a physics model (tube-bank convection, fin efficiency) against 12 wind-tunnel experiments; solved the problem with an exterior-penalty method and cyclic-coordinate and Hooke–Jeeves search, verified by multi-start runs and a grid search; now comparing with a neural-network surrogate.

Product Design & Development (MECH 525) | *Customer needs, benchmarking, prototyping* **Fall 2026**

- Five-person team taking a product from opportunity to a working prototype and a market pitch; screened 15 candidate opportunities with interviews, surveys, competitor and patent benchmarking and a weighted decision matrix.
- Converging on a low-cost 3D-scanning workflow for residual-limb capture and prosthetic socket design, targeting the access gap in prosthetic care (WHO: only 1 in 10 people who need prosthetic or orthotic devices have access). *(Ongoing.)*

NXP Cup Autonomous Racing Vehicle | *ROS 2, Gazebo, C++* **Feb 2025 – May 2025**

- Assembled the competition kit (sensors, actuators, control board, power) and built a modular ROS 2 stack with separate perception, localization, planning and actuation nodes.
- Optimized a minimum-time racing line under vehicle-dynamics constraints, validated in Gazebo and on track; joined inter-university training at Beirut Digital District.

Autonomous Irrigation System | *LabVIEW, myDAQ, Arduino* **May 2025**

- Led the design of a smart irrigation system using solar-radiation and soil-moisture sensing, with closed-loop control, real-time monitoring and manual/digital override in LabVIEW.

Microfluidic Mixing Analysis | *ANSYS, CNC, 3D Printing* **Jan 2025 – May 2025**

- Modeled two-inlet microfluidic mixing analytically (PDEs) and in CFD (ANSYS); fabricated CNC-machined micro-channels and validated them with dyed-fluid flow tests.

Go-Kart Design & Build | *SolidWorks FEA, Machining* **Jun 2025 – Aug 2025**

- Designed and validated a drive shaft for stress and fatigue in SolidWorks FEA; machined, assembled and tested a full-scale go-kart through iterative build–test cycles.

Leadership & Activities

AUB Robotics Club (leadership role, 2023 – present): organized competitive events (Build It 8.0); led an Arduino workshop series for 30+ students; mentor junior members on hardware, circuits and code.
Community service: 64+ hours per semester as a USAID scholar. **Model United Nations; Varsity sports:** represented AUB in Jiu-Jitsu and Taekwondo.

Skills

Control, estimation & optimization: state-space modeling, PID, LQR, observers, IMU calibration and Allan variance, constrained optimization
Robotics & autonomy: ROS 2 (Humble/Jazzy), Nav2, Gazebo, RTAB-Map / VI-SLAM, LiDAR-inertial SLAM, PX4/Pixhawk, NVIDIA Jetson, OptiTrack
AI & RL: Q-learning, DQN, dynamic programming, PyTorch, Gymnasium
Programming & tools: Python, C/C++, MATLAB/Simulink, Embedded C (ESP32, Arduino), Ubuntu, Git, LabVIEW
Design & manufacturing: SolidWorks (FEA), ANSYS, CNC machining, 3D printing
Languages: Arabic (native), English (fluent; IELTS Academic 8.0), French (elementary)